



Cafeteria Line Manager

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Problem Statement

The project simulates a cafeteria queue in a university.

*Students can arrive, leave, and be served.
The system tracks waiting time.*

The goal is to apply appropriate data structures and analyze time complexity.

Data Structures Used

ArrayDeque<String> → represents the queue

HashMap<String, Integer> → stores arrival times

int → current logical time

long → total waiting time

int → number of served students

Time Complexity Table

Operation	Complexity
ARRIVE	$O(1)$
VIP_ARRIVE	$O(1)$
SERVE	$O(1)$
PEEK	$O(1)$
LEAVE	$O(n)$
SIZE	$O(1)$

How the Program Works

Students enter commands in the console.

The program processes commands using loops and conditions.

The queue follows FIFO (First In, First Out).

VIP students are inserted at the front.

Waiting time is calculated using system time.

Empirical Timing Test

Used `System.nanoTime()` to measure performance.

Tested multiple runs.

Results show linear growth with number of operations.

Confirms theoretical complexity.

Example result:

100,000 insertions → consistent fast execution

Average time calculated over 5–10 runs

Why These Structures?

ArrayDeque is better than LinkedList for queue operations.

HashMap provides fast access to arrival time.

Using the correct data structure improves efficiency.

The project demonstrates asymptotic reasoning in practice.

Conclusion